

ECE 394 Lab Report 1

Control of Power Electronics

Tanya Nikam
TN22697

Luis Roman
LR1234

Alex Jang
ACJ2662

March 4, 2026



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1 Motor Characterization Experiments

In lab, we conducted various tests in order to investigate the physical characteristics of the bike and determine numerical parameters, gaining a better understanding of the electrical and mechanical system and what is required of us to be able to meet certain requirements. In this section, we detail which parameters are solved in each test, why the test was chosen, and how the parameters were obtained from the test result.

1.1 EMF Shape

To find the EMF shape, we can spin the wheel and measure the EMF produced with the oscilloscope. Doing so produces a graph like the one below:

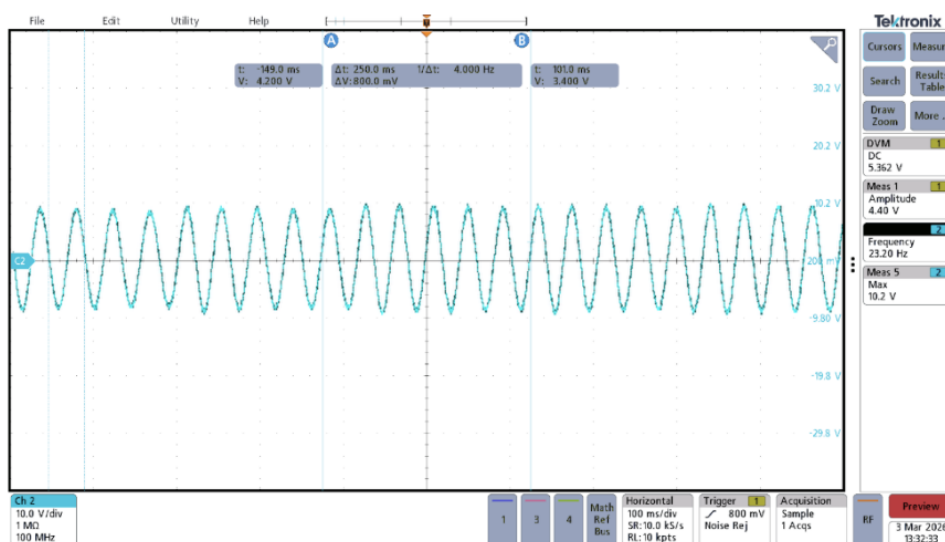


Figure 1: Flux Linkage

It seems that the EMF shape is sinusoidal, oscillating at a specific frequency. This makes sense, as the EMF should be related to the angular frequency of the wheel, which would create the periodic sine wave shown.

1.2 Number of Poles

To determine the number of poles, we looked at the EMF produced by one spin of the wheel. We know that each pole produces a peak and valley, aka one period of the waveform. By counting the total peaks produced by a single spin, we have just counted the number of poles around the wheel.

To make sure we weren't double counting poles, we filmed the oscilloscope data collection with the wheel spinning and made sure that we stopped the oscilloscope when we had done one full rotation. This produced the following screenshot:

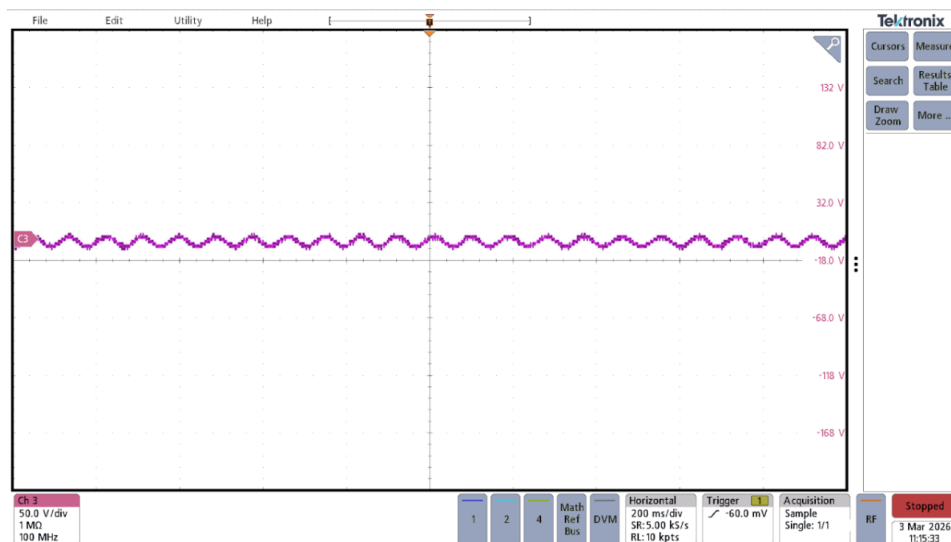


Figure 2: EMF Produced by One Spin of the Wheel

1.3 Winding DC Resistance r_w

To determine the DC resistance r_w , we directly measured this with the LCR meter. Below is a picture of the LCR meter output. This value was very unstable, so we approximated the resistance across two terminals as 395 mOhm.



Figure 3: LCR Meter Output

To find the DC resistance in each phase, we need to divide this value by 2, since we are measuring the resistance across both of the equivalent phase resistances. The final value would be 197.5 mOhm.

1.4 Winding Low-Frequency Inductance L_w

Similar to r_w , we directly measured L_w with the LCR meter. We determined that L_w was around 0.83 mH.

To find the total low frequency inductance per phase, we would need to divide this value by 2, since it would be measuring across both inductances. the final value would be 0.415 mH.

For both r_w and L_w , we were able to directly measure these quantities with the LCR meter because we eliminated flux linkage from the equation since we did not spin the wheel. Looking at the equation for a phase of the motor:

$$V_{AN} = ri_a + L \frac{di_a}{dt} + \lambda \omega_r \cos(\theta_r)$$

The V_{AN} is the sum of 3 different voltages. When we do not spin the wheel, the $\lambda \omega_r \cos(\theta_r)$ goes to 0. This means measuring the impedance across the terminals would only be measuring the r and L components.

1.5 Flux Linkage λ_m

To find flux linkage, we need to measure the EMF of the wheel from a specific angular frequency. We can then calculate flux linkage with this relationship:

$$\lambda_m = (\sqrt{3})(\text{amplitude})(1/\omega)$$

This was extrapolated from what we know about the relationship between the phase voltage and flux linkage, and how phase voltage is related to line voltage:

$$V_{AN} = r i_a + L \frac{di_a}{dt} + \lambda \omega_r \cos(\theta_r)$$

$$V_{AB} = \sqrt{3} V_{AN}$$

Our final flux linkage was calculated to be 0.038.

Angular frequency was determined by taking a video of the wheel, and counting the rotations per second. Amplitude was directly measured from the oscilloscope as seen in the screenshot below:

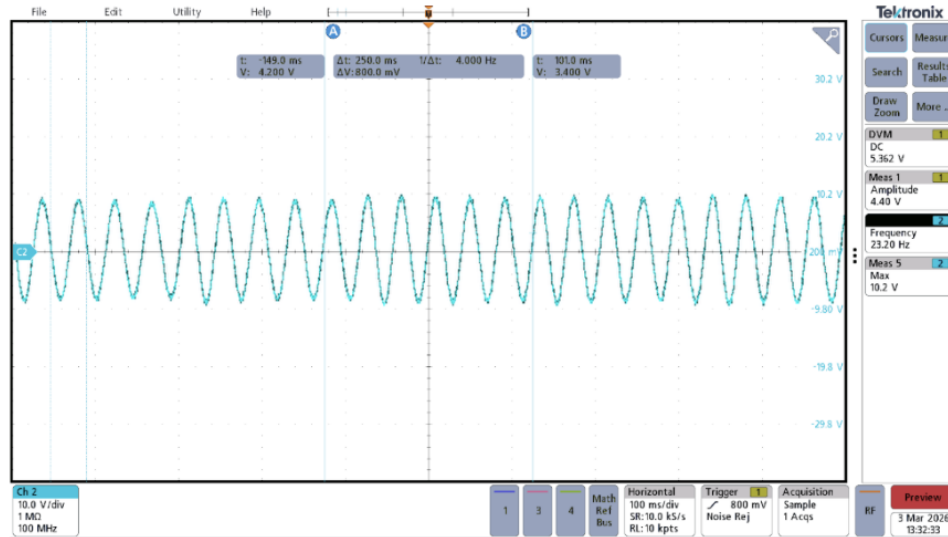


Figure 4: Flux Linkage

1.6 Relation between phases and hall sensor

To determine how the phases match up to each hall sensor, we measured both the hall sensor output and the corresponding phase EMF simultaneously. This was done for each of the three phases with a corresponding hall effect sensor.

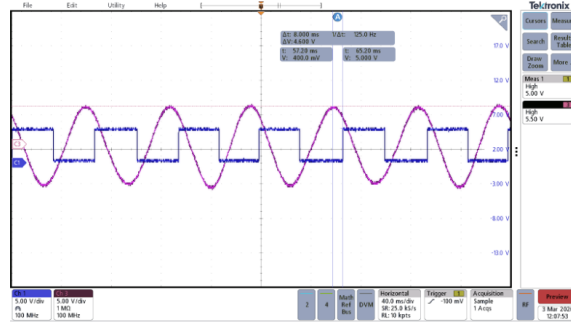


Figure 5: Hall Sensor and Phase A

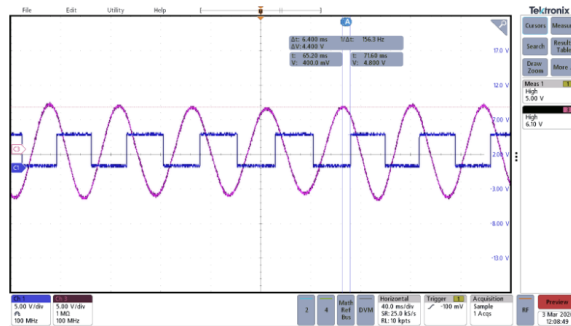


Figure 6: Hall Sensor and Phase B

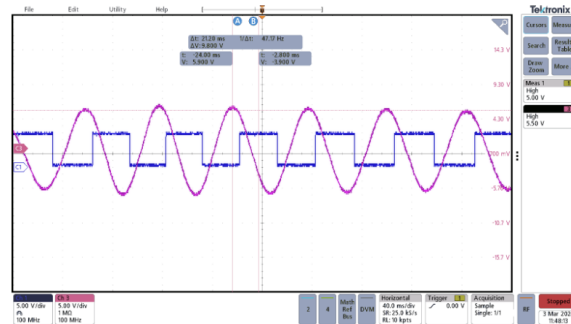


Figure 7: Hall Sensor and Phase C

It seems that each hall sensor is offset similarly from a corresponding phase EMF. By choosing to measure the phase EMF differently, we are able to align both waveforms. One example of this is shown below:

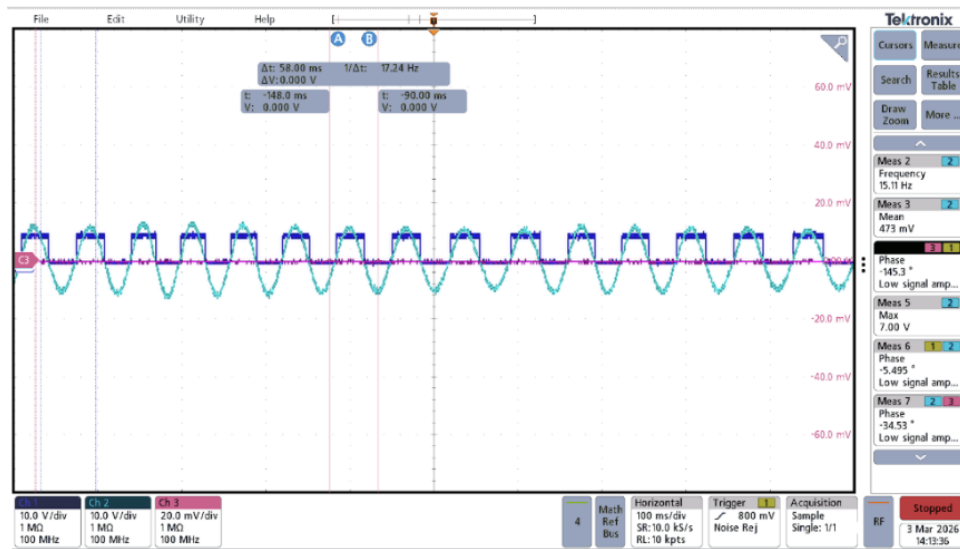


Figure 8: Hall Effect Sensor aligned to phase

This shows that each phase is aligned with the hall effect sensor. This makes sense because the hall sensor is measuring the magnetic fields that are induced from the effects of the corresponding phase branch rotating.

We found that the simulated waveforms for EMF and Hall Sensor output in PLECS matched our experimental results above. Each waveform is offset from the hall effect sensor by a similar amount, just like what we saw with the individual screenshots for phases A,B, and C.

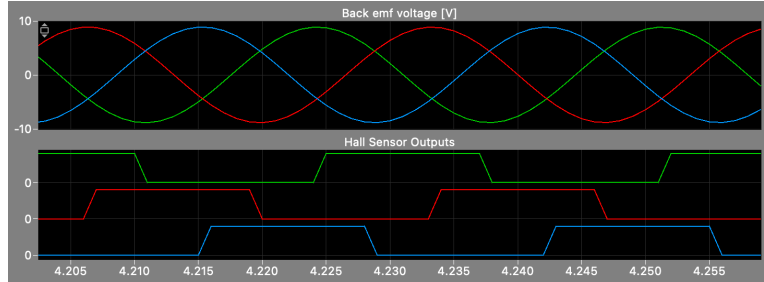


Figure 11: Simulation Layout on PLECS

2.2 Vehicle Dynamics

The specified performance metrics for our electric bicycle drivetrain are:

- Top speed of 10 mph or greater on flat road
- Ability to travel up a 3% grade ($\tan(\theta)$) at a speed of at least 5 mph

Use this section to include relevant code used for analysis, simulation, or control implementation.

Listing 1: Matlab Script for Calculating Physical Requirements

```

1 C_r = 0.013; % friction
2 M = 5.8 + 113.398; % bike + person (kg)
3 g = 9.81;
4 theta = atan(0.03);
5 C_d = 0.65; % drag
6 rho = 1.204; % air density
7 R = 350 * 10^-3; % radius of wheel (meters)
8 V = 4.4704; % speed (km)
9 A_v = 0.5; % area (m^2)
10 flux_linkage = 0.038;
11 poles = 46;
12 w_m = V/R;
13 T = C_r * M*g * R + M*g * R * sin(0) + 0.5*rho*A_v*C_d*V^2*R
14 P = T * V/R
15 back_emf = flux_linkage * P/2 * w_m
16
17 T_incline = C_r * M*g * cos(theta) * R + M*g * R * sin(theta)...
18 ... + 0.5*rho*A_v*C_d*(5)^2*R
19 P_incline = T * 5/R

```

What is the mechanical power required to meet both the top speed and gradeability requirements? (give one value)

To calculate the mechanical power required to meet these requirements we used Newton's Second law

$$J\dot{\omega} = \tau_e - \tau_m$$

Where the speed is constant and hence the previous expression is equal to 0.

To convert this to power, we apply the relationship between power and electrical torque:

$$P_e = \tau_e \omega_m$$

In the MATLAB script, we use speed and radius to calculate angular velocity, so the relationship looks more like this:

$$P_e = \tau_e (V/R)$$

Where V is speed and R is radius of the wheel.

The required power we calculated for the flat case was 85.4 W.

For the incline case we simply added gravity as a factor into the mechanical torque that is opposing the bike's movement. To quantize this added torque into the expressions we approximated this torque as acting on the outermost radius of the wheel for simplicity as the true gravitational torque would act on the center of mass of the bike/rider system. Hence the added torque term added to the total mechanical torque has the form

$$\tau_g = MgR \sin \theta$$

After recalculating the required torque for the desired steady state, we obtained that the required power to maintain 5mph at 3% gradient is 95.6W.

What is the peak back-emf voltage that will be generated at a speed of 10 mph?

To calculate peak EMF we used the following equation

$$e = \lambda \cdot \frac{P}{2} \cdot \omega_m$$

Using the parameters in PLECS, alongside the angular velocity for 10 mph, we can calculate peak emf. We got 9.024V.

Below is a screenshot of the PLECS speed simulation.

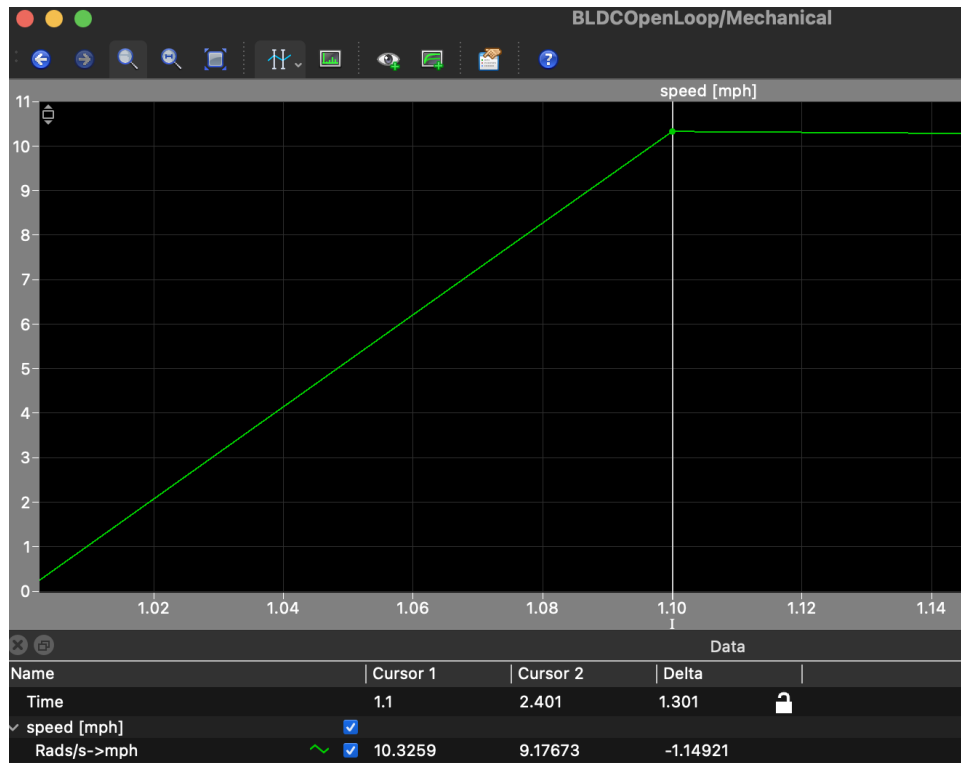


Figure 12: Simulation at peak speed of 10 mph

Consider and discuss, briefly, how each of these will inform the design of the power electronics in the ensuing experiments.

Understanding the mechanical power and back EMF produced by the motor to meet the requirements is important in implementing power electronics controls. The mechanical power is directly related to electrical torque, which is proportional to i_q and the back EMF. With both back EMF and electrical torque, we can find i_q .

This allows us to implement control systems we have already developed for simple converters in the dq domain for 3 phase motors.